

Planning in E2E Autonomous Driving Systems

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Introduction

Levels of Driving Automation

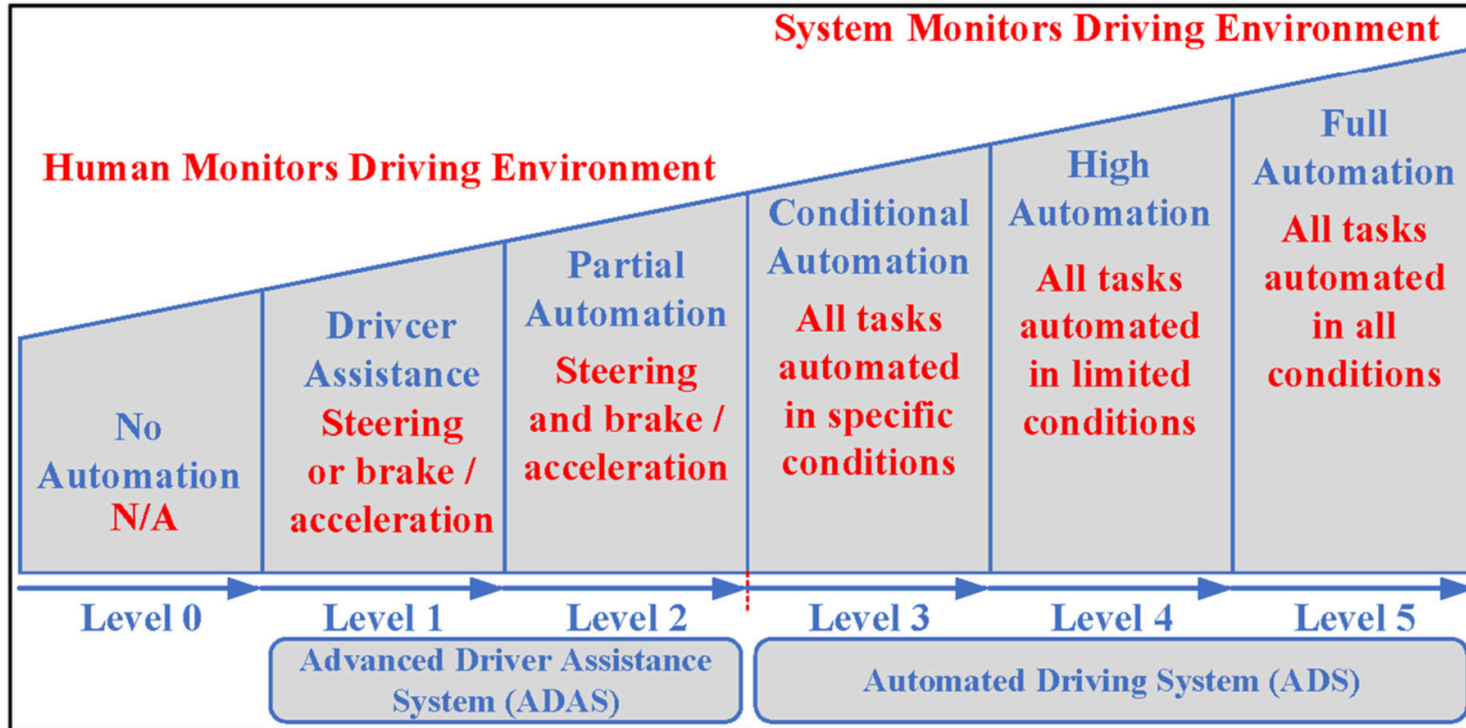


Figure 1: Levels of Driving Automation [1]

Why is automation in driving required?

- Reducing road accidents
- Decreasing traffic congestion
- Minimizing fuel consumption
- Reducing air pollution

- There is a vast amount data available for the autonomous driving scenario along with highly competitive research.
- This can lead to high quality models which can be fine-tuned for other applications like [Robotics](#) and [UAVs](#)¹

¹Unmanned Aerial Vehicles

Current Scenario of Autonomous Driving

- To date very few models of private cars equipped with automation level 3 are legally allowed to circulate with limitations on location and speed [1]
- While no private vehicle is licensed to roam public roads in “full” or “semi” autonomous mode (i.e., automation level 4 and 5) [1]

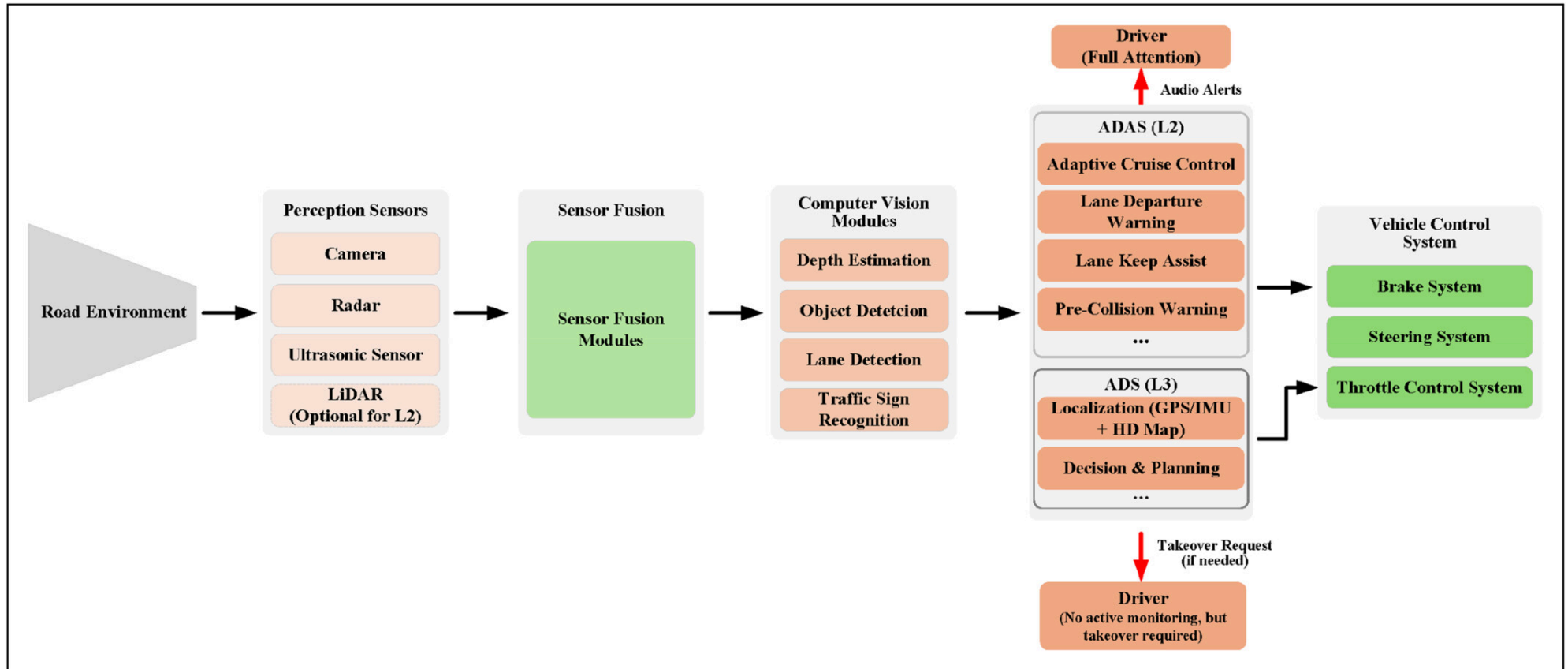


Figure 2: Schematic overview of ADS perception pipeline [1]

Challenges / Problems

- **Perception:**
 - sensor limitations,
 - depth estimation,
 - object detection,
 - sun glare,
 - adverse weather,
 - data size,
 - real-time localization and mapping
- **Path Planning:**
 - Real-time decision making,
 - handling uncertainty and failures,
 - safety vs efficiency,
 - behaviour prediction,
 - automated negotiation

Challenges / Problems

- **Common-sense Reasoning:**
 - Grounding problem,
 - Cornercase scenarios,
 - accuracy and safety requirements,
 - vulnerable road users,
 - Narrow vs General AI
- **Road Infrastructure:**
 - Inconsistent road signs,
 - unpredictable road conditions
- **Ethics:**
 - Trolley Problem,
 - Accountability

Challenges / Problems

- **End-users:**
 - Situational awareness,
 - explainable automation,
 - data privacy
- **Others:**
 - Policy making
 - ...

Why E2E (end-to-end) models?

- Traditional systems use a sequential pipeline of task specific modules leading to
 - **information loss** due to data abstraction at each module
 - **cascading errors** as downstream tasks cannot correct upstream mistakes

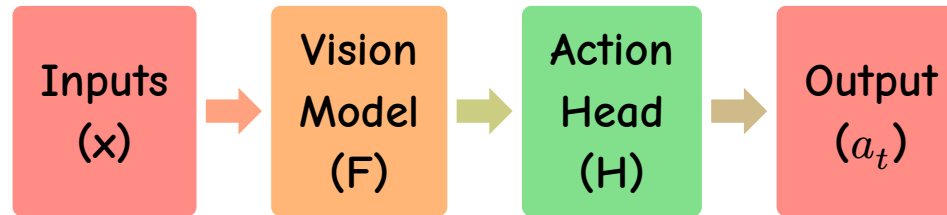


- E2E models primarily address these fundamental structural limitations of traditional pipelines along with **unified objective** and **reduced engineering complexity**

End-to-End Models

Problem Formulation

$$a_t = H(F(x|\theta))$$



Inputs (x)	Backbone (F)	Outputs (a _t)
• Sensor observations • Latent scene representations • Language instructions • Proprioceptive states	• VLM/VM as backbone	• Trajectory (discrete/continuous) • Direct Control • Language
	Action Head (H) • Predicts next action	

Inputs (x)

- Sensor Inputs

- Visual Images: $x_{\text{img}} \in \mathbb{R}^{N_C \times H \times W \times 3}$
- LiDAR Point Clouds : $x_{\text{lidar}} \in \mathbb{R}^{N_p \times D}$

- Latent Representations

- Bird's-Eye View (BEV) Features: $x_{\text{bev}} \in \mathbb{R}^{C \times H_{\text{bev}} \times W_{\text{bev}}}$
- Occupancy Grids: $x_{\text{occ}} \in \mathbb{R}^{C_{\text{occ}} \times X \times Y \times Z}$

- Language Inputs: $x_{\text{lang}} \in \mathbb{R}^{T \times D_{\text{emb}}}$

- Ego-Vehicle State Information: $x_{\text{state}} \in \mathbb{R}^{D_{\text{state}}}$

VLM/VM as Backbone (F)

- VLM/VM for direct action generation (Single System)
- VLM for guidance generation (Dual System) (slow thinking + fast execution)

Action Heads (H)

It converts the latent representation from VLM to action outputs

- **Language Head:** generates free-form textual commands which are then passed to a parser for final actions
- **Regression Head:** directly predicts continuous numerical values like steering angles, throttle and brake values
- **Trajectory Selection:** Evaluates a set of candidate trajectories and selects the most optimal one
- **Trajectory Generation:** Generates trajectory through probabilistic modelling of actions

Outputs (a_t)

- Discrete Trajectory Representations¹

$$a_t = \{(x_i, y_i)\}_{i=1}^{\varphi}, \text{ where } (x_i, y_i) \in \mathbb{R}^2$$

- Continuous Trajectory Representations²

$$a_t = (v(t), k(t)), \quad \text{for } t \in [0, T]$$

- Direct Control Representations³

$$a_t = (\delta_t, \tau_t, \beta_t)$$

- Language Representations⁴

$$a_t = \{w_1, w_2, \dots, w_T\}$$

¹ $(x_i, y_i) \rightarrow$ coordinates, $\varphi \rightarrow$ prediction horizon

² $v(t) \rightarrow$ speed profile, $k(t) \rightarrow$ curvature profile, $T \rightarrow$ time horizon

³ $\delta_t \rightarrow$ steering angle, $\tau_t \rightarrow$ throttle input, $\beta_t \rightarrow$ brake input

⁴ $w_i \rightarrow i^{\text{th}}$ token in the models vocabulary $\mathcal{V}$

Vision-Action Models

End-to-End Models

Directly predicts the control actions or the trajectories from sensor inputs (*action as output*)

- Action-Only Models
- Perception-Action Models

World Models

Explicitly models action-conditioned future dynamics to support policy learning and decision making (*action as input*)

- Image-Based World Models¹
- Occupancy-Based World Models²
- Latent-Based World Models

(Generally Diffusion-Based or Autoregressive Models)

¹outputs: raw pixels

²outputs: free space geometry

Vision-Action Models

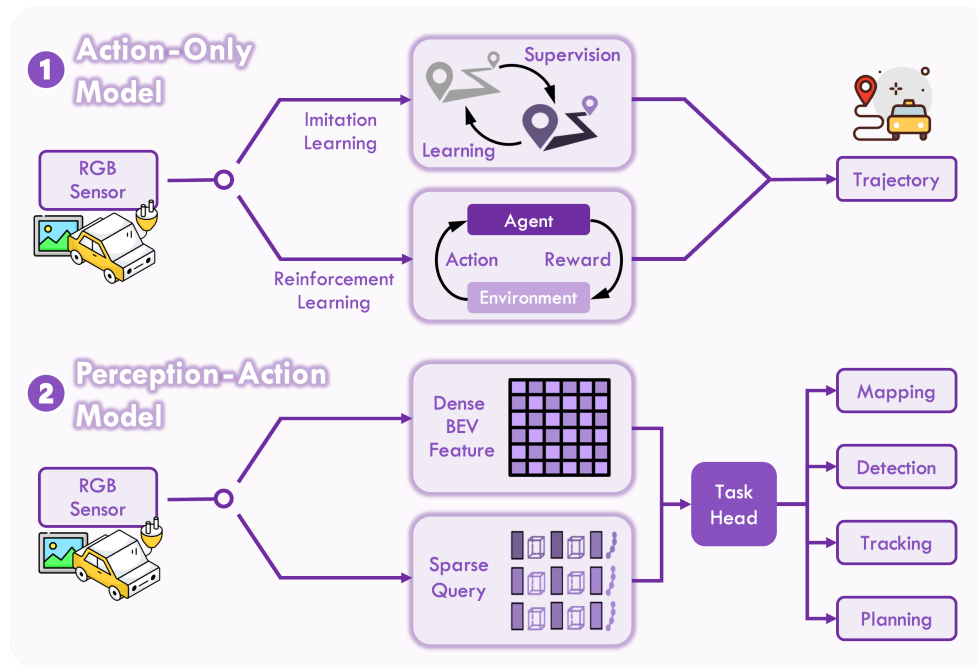


Figure 3: End-to-End Models

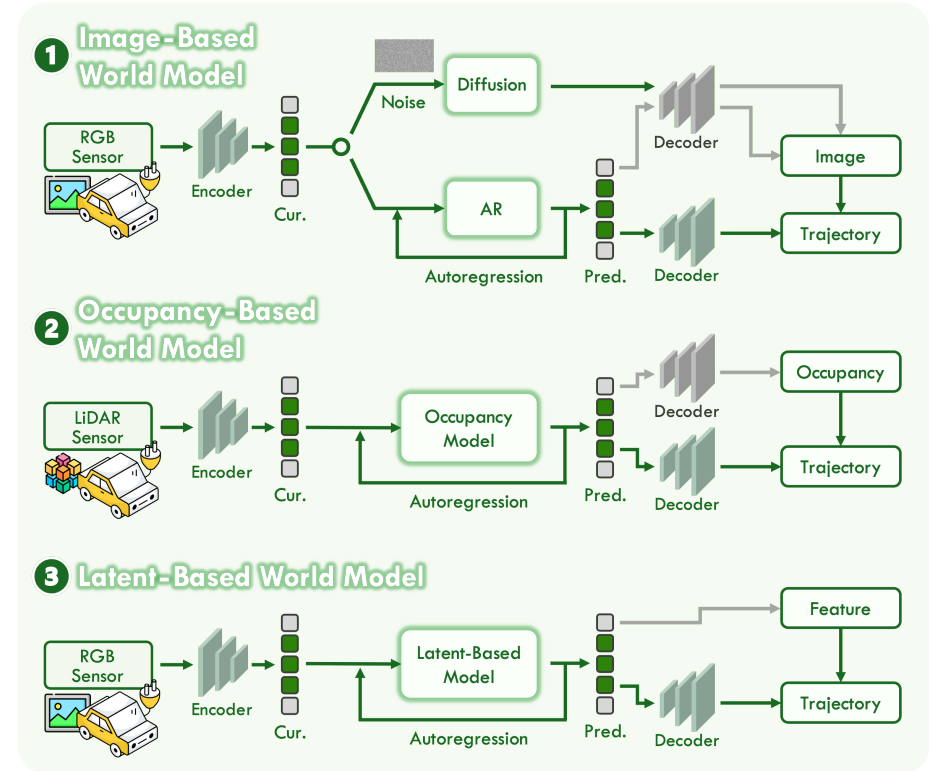


Figure 4: World Models

Why aren't Vision-Action Models sufficient?

Vision-Action (VA) Models

- Limited interpretability
- Fragile generalization
- Directly map perception to low-level actions

Vision-Language-Action (VLA) Models

- VLM backbone with action-prediction head
- Instruction following

Vision-Language-Action Models

End-to-End VLA

Single model that directly maps the inputs to actions

- Textual Action Generator
 - Meta Actions
 - Trajectory Waypoints
- Numerical Action Generator
 - Additional Action Head
 - Additional Action Tokens

Dual-System VLA

Dual model, a slow planner with a fast executor

- Explicit Action Guidance
 - Meta Action Guidance
 - Waypoint Supervision
- Implicit Representations Transfer
 - Knowledge Distillation
 - Multimodal Feature Fusion

Vision-Language-Action Models

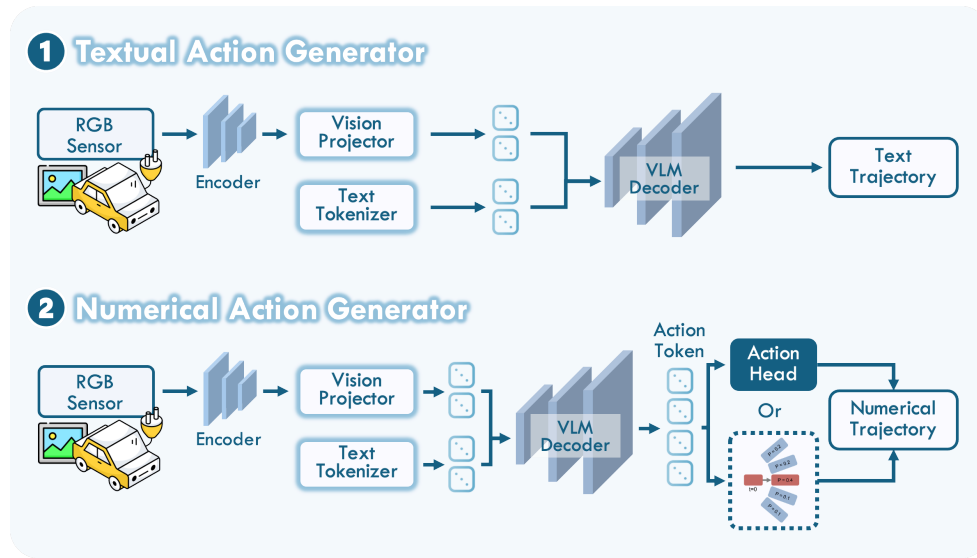


Figure 5: End-to-End VLA

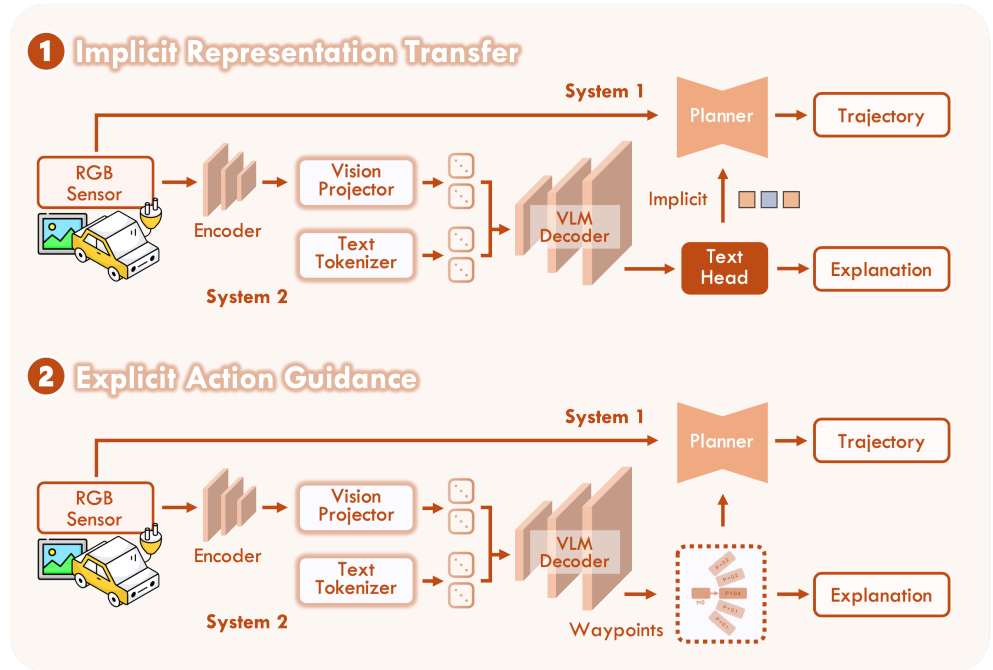


Figure 6: Dual-System VLA

VLA: Challenges

- Real-Time Processing and Latency
- Lack of Domain-Specific Foundational Models
- Generalizing to Rare and Novel Scenarios
- High-Quality Data Cost
- Interpretability and Hallucination
- Long-Horizon Temporal Coherence

Future Directions

- Unified Vision-Language World Models
- Richer Multimodal Fusion
- Socially Aware, Knowledge-Grounded Driving
- Continual & Onboard Learning
- Standardized Evaluation & Safety Guarantees
- Human-Centric Interaction & Personalization

Experiments

Datasets

Most widely used datasets

Open-Loop dataset, uses trajectory-based metrics	nuScenes [2]
Open-Loop dataset, uses 3D Gaussian Splatting for pseudo-simulation of closed-loop dataset	NAVSIM [3]
Closed-Loop dataset, built on CARLA [4] (simulator)	Bench2Drive [5]
Most recent dataset introducing long-tail, safety-critical scenes with human-preference annotations	WOD-E2E [6]

Evaluation Metrics

Action-Planning Open-Loop Evaluation

L2 ↓	L2 Error	L2 distance error between the planned trajectory and the human driving trajectory in 3 seconds
CR ↓	Collision Rate	How often the self-driving vehicle would collide with other agents on the road
ADE ↓	Average Displacement Error	Mean displacement error between predicted trajectories and expert waypoints across the horizon, reflecting overall trajectory accuracy
FDE ↓	Final Displacement Error	Displacement error at the final predicted waypoint compared with expert trajectories, emphasizing long-term accuracy
MR ↓	Miss Rate	Fraction of prediction time steps where displacement error exceeds horizon-specific thresholds, reflecting failure in trajectory coverage
AHE ↓	Average Heading Error	Mean absolute angular deviation between predicted and expert heading over the trajectory horizon, measuring orientation accuracy
FHE ↓	Final Heading Error	Absolute angular deviation of predicted heading from expert at the final timestep, reflecting terminal orientation accuracy
SLE ↓	Speed L1 Error	Mean absolute error of predicted speed control signals
SALE ↓	Steer Angle L1 Error	Mean absolute error of predicted steering angle control signals

Trajectory-Based Closed-Loop Evaluation

RC ↑	Route Completion	The percentage of route distance completed
DS ↑	Driving Score	RC weighted by a penalty factor that accounts for collisions with pedestrians, vehicles, etc
NC ↑	No Collision	Fraction of scenarios without ego-fault collisions, focusing exclusively on responsibility-aware collision evaluation
DAC ↑	Driving Admissibility Check	Boolean evaluation that checks whether the ego vehicle remains inside drivable polygons throughout the rollout
TTC ↑	Time To Collision	Boolean verification that the time-to-collision value exceeds safety thresholds, preventing imminent crashes
C ↑	Driving Comfort	The comfort of driving
EP ↑	Ego Progress	Penalization of excessive jerk, acceleration, or yaw-rate, reflecting ride quality and passenger comfort
PDMS ↑	Predictive Driver Model Score	A flexible weighted evaluation score in autonomous driving that aggregates multiple safety, progress, and comfort subscores into a single metric
SR ↑	Success Rate	Percentage of navigation episodes that successfully reach the goal within a fixed time budget, indicating overall task completion
ID ↑	Infraction Distance	Average driving distance between two infractions, with longer distances reflecting safer and more reliable policy behavior

Text-Based Action Evaluation

CIDEr ↑	Consensus-based Image Description Evaluation	Measures similarity of generated captions to multiple human references using TF-IDF weighted n-grams
BLEU ↑	Bilingual Evaluation Understudy	Precision-based metric that compares n-grams of the generated text against reference texts
METEOR ↑	Metric for Evaluation of Translation with Explicit Ordering	Considers unigram precision and recall with stemming, synonym matching, and fragmentation penalty
Rouge ↑	Recall-Oriented Understudy for Gisting Evaluation	Recall-focused metric using overlapping n-grams, word sequences, or word pairs between generated and reference texts
Top-1 Acc ↑	Visual Question Answering Top-1 Accuracy	Percentage of predictions where the most confident output matches the ground truth label

References

Important Papers

1. Why Autonomous Vehicles Are Not Ready Yet [1]
2. Vision-Language-Action Models for Autonomous Driving [7]

Others

- **Awesome VLA Models:** <https://github.com/worldbench/awesome-vla-for-ad>
- **Github Repository:** <https://github.com/airexlab/AutonomousSystems>



Figure 8: Our Github Repository: <https://github.com/airexlab/AutonomousSystems>

Bibliography

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- [6] R. Xu *et al.*, “WOD-E2E: Waymo Open Dataset for End-to-End Driving in Challenging Long-Tail Scenarios,” *arXiv preprint arXiv:2510.26125*, 2025.
- [7] T. Hu *et al.*, “Vision-Language-Action Models for Autonomous Driving: Past, Present, and Future.” [Online]. Available: <http://arxiv.org/abs/2512.16760>